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The revolution of Pure Mathematics on a Field basis

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Abstract

The paper analyzes the extent to which the principle of least action can be considered to have prompted a scientific revolution, as defined by the historian and philosopher of science Thomas Kuhn. It traces back the development of the principle of least action, from its very beginnings in the 18th century to solve problems regarding optics to the subsequent development of Lagrangian and Hamiltonian mechanics, and attempts to align its rise with the suggested stages of scientific revolutions: normal science, crisis, revolutionary science, paradigm shift, and new normal science. It is found that the discovery of the principle aligns somewhat with Kuhn's definition, in what one could call a modified view of the Kuhnian scientific revolution. The paper furthermore showcases the consistency of the principle of least action with newer theories in modern physics, including quantum mechanics and general relativity, and its applicability to several other branches of physics, solidifying its status as a lasting revolution in our understanding of physics. In sum, the paper aims to shed light on the many existing and prospective applications of the least action principle and its role in furthering our understanding of reality.

Keywords: *principle of least action, scientific revolution, Kuhnian revolution, quantum mechanics, general relativity, history of science*

Introduction:

Not all scientific discoveries are created equal. It is those that come close to the 'theory of everything' that garner the most attention: naturally, for much of the history of scientific discovery, particularly in the field of physics, has been concerned with finding simple, yet universal principles. Humanity has not yet reached a 'theory of everything', of course, yet these theories occasionally prompt scientific revolutions that significantly further our understanding of their respective fields.

This paper aims to evaluate this claim in the context of the Principle of Least Action (PLA), a principle first developed by Pierre Louis Maupertuis in the 18th century that has developed to encompass nearly all of modern physics. Many definitions of the scientific revolution have been proposed, and this paper takes on that of Thomas Kuhn, as elucidated in his book *The Structure of Scientific Revolutions*, and identifies the extent to which the PLA could be said to have successfully brought forth such a revolution through considering the historical context behind the emergence of the theory, its development, and its subsequent effects on the field of physics. The paper furthermore showcases how this principle fits into our current understanding of reality and the many applications it has in various disciplines of physics.

What is a Scientific Revolution?

The existence and nature of scientific revolutions is heavily debated, chiefly among historians and philosophers of science. The topic remains controversial, with many differing, sometimes contrasting, definitions of scientific revolutions. For instance, in 1949, political historian Herbert Butterfield argued in his book *The Origins of Modern Science: 1300–1800* that a scientific revolution is a major event that results in the creation of a new scientific field. As is clear from the title of the book, Butterfield was concerned only with origins, not later developments (Internet Encyclopedia of Philosophy). Thomas Kuhn and Paul Feyerabend later brought new perspectives to the philosophy of science by emphasizing the importance of revolutionary changes and the idea that scientific revolutions are ongoing, not just historical events. Kuhn argued that revolutions do not only create new fields, they extend and develop the field further (although Kuhn eventually abandoned Butterfield's definition of formative scientific revolutions).

Later in 1985, in his book *Revolution in Science*, I. Bernard Cohen listed four general criteria that would serve to identify a true scientific revolution: the scientists involved must see themselves as revolutionaries, and other relevant individuals must agree that a revolution is underway; historical documents must label it a revolution; later historians and philosophers must recognize it as such; and future scientists in the field must agree (Cohen and Schreiber). This view places great emphasis on conscious awareness: scientists who work on possibly groundbreaking discoveries yet who do not consider themselves revolutionary would not be considered to have brought forth any revolution, and perhaps reflects a more political, rather than scientific, definition of a revolution (Ruark). It also assumes a very linear progress in science, which is very much not the case, more often than not.

The Kuhnian Revolution and Criticisms

Of the most well-known definitions of a scientific revolution is that of Thomas Kuhn, as elucidated in his book *The Structure of Scientific Revolutions*, wherein he writes "scientific revolutions are here taken to be those non-cumulative developmental episodes in which an older paradigm is replaced in whole or in part by an incompatible new one" (Kuhn 311).

In Kuhn's view, a scientific field matures when a specific set of problem-solving methods becomes established, forming a "paradigm" that defines what constitutes legitimate scientific work. This paradigm dictates what is considered "normal science" and directs researchers toward solving problems within its framework. Normal science, by nature, is focused and incremental, encouraging research within established boundaries while discouraging revolutionary ideas or discoveries that might challenge the prevailing paradigm (Stanford Encyclopedia of Philosophy). However, as research progresses, anomalies – results that do not fit within the existing paradigm – are inevitable. When these anomalies accumulate and cannot be resolved by the current paradigm, the scientific community may enter a crisis period. During this time, alternative theories and solutions are considered. A new paradigm that offers a more compelling explanation can lead to a paradigm shift, or a Kuhnian revolution. This shift occurs when the new paradigm replaces the old one, introducing a new set of research problems and methods and ultimately rendering the previous framework outdated or flawed. Revolutions consistent with this definition include the Copernican revolution of the 16th century and Antoine Lavoisier's 18th-century chemical revolution.

Despite the prominence of Kuhn's theory, it is not by any means entirely true, and has suggested many controversial ideas that have been widely criticized. For instance, Kuhn presents new paradigms as completely distinct from older ones, neglecting the influence of the latter on the former (Athens). He also claimed that paradigms are completely incommensurable, a view which many thought was too radical. In response to this claim, Carl Kordig argued that, despite paradigm differences, rival scientific theories share some observations and meanings, allowing for meaningful comparison and communication on the observational level (Kordig).

In light of the aforementioned, Kuhn's definition of a scientific revolution will be utilized not out of an assumption that it is correct, but due to its relative clarity and utility in testing the proposed hypothesis, for, despite its shortcomings, the theory remains valuable in understanding the nature of scientific progress.

The Principle of Least Action

The principle of least action was likely first formulated in the 18th century by Pierre Louis de Maupertuis, in relation to optics (or geometry, in some notations). It began as a metaphysical idea, proposing that nature operates by minimizing a quantity called "action," which Maupertuis defined as the product of mass, velocity, and distance over a path. Maupertuis believed this minimization reflected nature's economy and used it to explain phenomena like the path of light during refraction (Sources). Maupertuis did not provide much mathematical proof of his theory and instead relied on more metaphysical and teleological underpinnings for his principle, which resulted in it receiving heavy criticism in its early days (Voltaire; Serafini). Voltaire satirized Maupertuis's assertion that nature always acts in the most economical way as overly speculative and pretentious, and viewed Maupertuis's reliance on metaphysics and appeals to God's design as unscientific and lacking empirical support (Brown).

Around the same time, Leonhard Euler independently formulated a similar principle and gave it a more mathematical foundation, connecting it to the calculus of variations and applying it to broader mechanical systems (in cases of constant energy) (Euler). Joseph-Louis Lagrange and William Rowan Hamilton transformed the principle by defining action as the time integral of the difference between kinetic and potential energy (the Lagrangian), and showing that a system's path keeps the action stationary, but not necessarily minimal. Hamilton furthermore connected the principle to that of Fermat, and consequently arrived at what is now known as Hamilton's Principle (Sources). This evolution transformed Maupertuis's original, somewhat metaphysical idea into the rigorous, general "principle of least action" foundational in modern physics.

The Principle of Least Action as a Kuhnian Scientific Revolution:

Upon defining Kuhnian scientific revolutions and the historical context behind the discovery of the principle of least action, it becomes possible to assess the extent to which the principle could be said to have brought forth such a revolution. To do so, in the most elementary sense, would involve aligning the process of the discovery of the principle to the stages of a Kuhnian scientific revolution.

Prior to the principle of least action, scientists operated under the framework of Newtonian mechanics. This represents the dominant paradigm in the initial period of "normal science." Over time, however, physicists began encountering increasing difficulties when dealing with complex systems, particularly in relation to the paths and motion of particles. These unresolved anomalies constituted the "crisis," as they called into question the completeness and universality of the existing paradigm.

Then, the principle of least action was proposed by Pierre Louis de Maupertuis and subsequently reinforced by Leonhard Euler and Joseph-Louis Lagrange through rigorous mathematical formulations. The principle accurately predicted the motion of particles, resolving the anomalies, and thus successfully constitutes a period of "normal science."

This subsequently resulted in a sort of paradigm shift, albeit not a complete one. Unlike the classic Kuhnian revolution, the principle of least action did not overthrow or render obsolete the existing paradigm (i.e., Newtonian mechanics). Instead, it provided an alternative, more fundamental formulation that could reproduce Newton's laws as a special case. The equations of motion derived from the principle of least action are mathematically equivalent to those of Newtonian mechanics. Therefore, in the following period of normal science, the principle of least action and Newtonian mechanics coexisted. This coexistence means that the adoption of the action principle did not involve the wholesale replacement of one paradigm with another, as Kuhn described in his model of scientific revolutions, but rather an expansion and deepening of the conceptual foundations of physics, which consequently opened more doors for scientific discovery.

Applications and Implications of the *Principle of Least Action*:

The principle of least action can be used to underlie all of physics: it can be used to derive not only the laws of motion but also Maxwell's equations and Fermat's principle, corresponding to electromagnetism and optics, respectively (Ross and Roche).

Notably enough, the principle of least action finds one of its most notable applications in quantum mechanics, despite being formulated centuries earlier, particularly through the Feynman path integral formulation. This approach states that a quantum system does not follow a single path but rather explores all possible paths between two points (Feynman). Each path contributes to the system's probability amplitude, weighted by an exponential of the action (in units of Planck's constant). The classical path, where the action is stationary, dominates in the limit where quantum effects become negligible, thus recovering classical mechanics. In this way, the principle of least action can be seen as unifying classical and quantum mechanics.

Furthermore, both special and general relativity use the principle to describe the motion of bodies and the curvature of spacetime, and it can be used to derive Einstein's field equations. The action in general relativity is defined by the Einstein-Hilbert action, which involves integrating the scalar curvature (a measure of spacetime curvature) over the entire spacetime volume (Misner 207).

Conclusion:

This paper has demonstrated that the discovery and development of the principle of least action at the hands of Pierre Louis de Maupertuis, Leonhard Euler, Joseph-Louis Lagrange, and William Rowan Hamilton can in fact be considered a scientific revolution in the Kuhnian sense. The principle of least action is now known as a unifying idea in physics, and the ability of unifying many disparate ideas in a large field is, in both the literal and the Kuhnian sense, revolutionary.